

H2GUARD · DESIGN DECISION RECORD

H2Guard recommendations accepted

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DOCUMENT CONTROL

Document ID	HGD-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/h2guard		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	02 to 07 decided by Amish on 2026-10-02 (recommendations approved, HGD-DEC-001)
0.1	2026-09-25	Amish Chadha	Record the review recommendations accepted by Amish on 2026-09-25 and what changed in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below with a recommendation is decided by Amish, 2026-09-25: go with recommendation. Items without a recommendation (O2 to O7) were decided on 2026-10-02, when Amish approved the recommendations later written for them ("i approve your recommendations for all 555 open decisions."); see Table 2 and HGD-DEC-001.

■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." The items concerned are those in `docs/REVIEW.md` (TRL 2 and TRL 3 sessions) and in HGD-DDR-001 that were marked "Proposed, awaiting Amish" or "Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review" and that carried a recommendation. Where a recommendation offered several options, the recommended option is the decision. Work stays at TRL 3; TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are those listed in `docs/REVIEW.md` and HGD-DDR-001.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
1	D1 to D9 of HGD-DDR-001 (sensor pair, set points, normally closed valve, continuous ventilation with boost, inventory rule, key reset and one head per room, university teaching lab as first users, independent hardware trip, 24 V DC)	As recommended	Status wording in HGD-DDR-001, HGD-PRB-001, HGD-PRC-001 and HGD-REQ-001; no design change
2	Budget (TRL 2 item 1 and HGD-DDR-001 O1)	Option (a) of the TRL 3 review: \$265, covering the TRL 3 kit including the fan and make-up grille. The TRL 2 figure of \$250 was superseded by this later recommendation	<code>budget_usd</code> 180 to 265; R14 target \$265; R14 from not met (\$264 against \$180, +47 %) to met on paper (\$1 margin)
3	Timed escalation from warning to trip	Option (a): close the valve when a warning is held for 5 min, latched as a trip	R3 restated; firmware rule added to HGD-PRC-001; HGD-CAL-001 v0.2 F7 and F8: off-axis design leak closed at 5.6 min, 28 L released, room about 1.9 % LFL; GA note on HGD-DWG-001 Rev P2
4	Head placement rule	Adopt: head directly above each likely leak point, second head where leak points are far apart; the horizontal offset limit is set at TRL 4	R1 restated; HGD-CAL-001 E7 gives the plume radius at the ports (141 mm) as the paper basis; GA note on HGD-DWG-001 Rev P2. Setting the offset limit by test is TRL 4 work, decided but on hold
5	R13 scope	Option (b): the fan and grille wall openings are builder's work outside the 4 h target	R13 restated; status from not met (6.0 h) to at risk (4.1 h, 5 min over)
6	H2Bench to reflect the inventory rule (D5)	Adopt	Cross-repo action listed in <code>docs/REVIEW.md</code> ; H2Bench not edited

No pitch or problem rewording was recommended, so `project.yaml` and the README keep their pitch and problem text. No part was added or resized, so the model geometry, STEP and STL files and the BOM lines are unchanged; the drawing changes only in its notes.

Table 2. Items left open here (no recommendation was made then), decided on 2026-10-02.

#	Item	Decided by Amish, 2026-10-02 (HGD-DEC-001)
O2	Oxygen depletion monitoring when inert gas cylinders share the room	Oxygen sensor added, as a standard kit option, whenever inert gas cylinders share the room
O3	Where the set points and the inventory rule come from in each country the first users are in	Stricter of the national code and ISO 26142, and the host institution's safety office where stricter; 10 % and 25 % LFL stay the defaults
O4	How airflow is proven: tachometer only, or a differential pressure switch	Differential pressure switch at the fan
O5	Low-cost solenoid valve or a certified gas valve	Valve rated for hydrogen with a stated seat leak rate; the brass valve only with the maker's hydrogen and seat leak ratings and a seat leak proof test at TRL 4
O6	Whether a full-span bump test trips the system (proof test with a key reset) or a lower span gas is used; no recommendation until the cup delivery factor is known	The 1 % full-span bump test trips the system; the pass level is set once the cup delivery factor is measured
O7	A second interlock output (dry contact) for H2Bench's bench supply; raised as an interface point, with no recommendation	Dry-contact relay output that opens on a trip or loss of power

■ Consequences

- Requirement status (HGD-CAL-001 v0.2): 0 not met, 3 at risk (R4, R7, R13), 1 not verifiable at TRL 3 (R1), 7 met on paper, 4 met by design. Before: 2 not met, 2 at risk, 1 not verifiable, 6 met on paper, 4 met by design.
- Documents revised: HGD-PRB-001 v0.4, HGD-PRC-001 v0.4, HGD-REQ-001 v0.4, HGD-CAL-001 v0.2, HGD-DDR-001 v0.2, HGD-DWG-001 Rev P2.
- The timed escalation is a firmware rule and does not share the independence of the comparator trip (R6). Writing that firmware beyond a labeled sketch is TRL 4 work and is on hold.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building, testing, purchasing or firmware work.